T42-Flood Detection Using Sentinel-1 SAR and Sentinel-2 Satellite Data

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Objective

Developed an automated system to detect floods using multi-temporal Sentinel-1 SAR and Sentinel-2 optical/NDVI data. SAR enables reliable monitoring even under cloud cover, improving accuracy of flood mapping.

Implementation

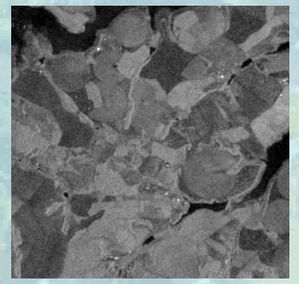
1. Data Acquisition: Collect SAR (VV, VH) + Optical + NDVI images (Before & After)
2. Preprocessing: dB conversion, noise filtering, normalization
3. Fusion: Combine multi-modal channels
4. Model: U-Net extracts temporal changes
5. Post-Processing: Binary flood mask,area calculation

Contributions

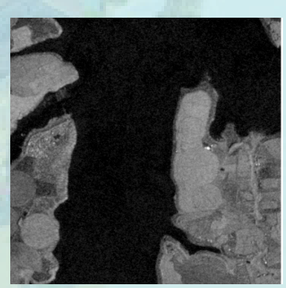
1. Combines SAR + Optical data for robust detection
2. Works under cloudy conditions
3. Detects both flood extent and presence
4. Uses deep learning for accurate mapping

Result

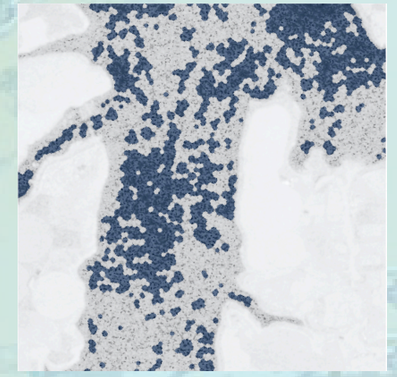
before



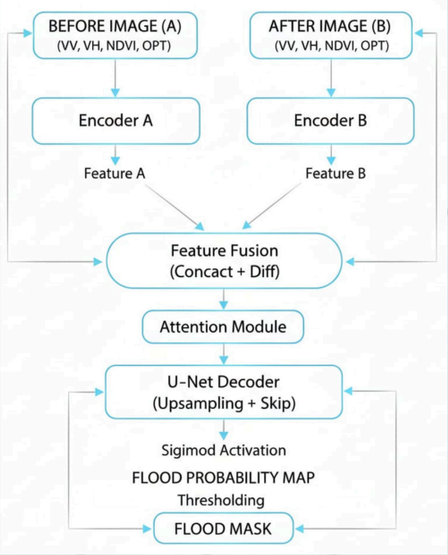
after



Prediction



Design



Metric

Accuracy	0.9377
Precision	0.8217
Recall	0.9060
F1-Score	0.8618
IoU(Jaccard)	0.7571

Conclusion

The proposed system successfully detects flooded regions using multi-modal satellite data and deep learning. By combining SAR and optical imagery, the model achieves reliable performance even under cloud cover.

Team

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